

Sales Impact of
Layout Change:
Evidence from
Matched Control
Stores



Executive Summary



Question

Does the new store layout improve store performance?



Key Findings

- Store 77 shows the strongest evidence of a positive intervention effect in both total sales and customer count during the trial period.
- Analysis suggests that the new layout has the potential to improve store performance, but the effect is not uniform across locations.



Recommendations

- A broader rollout should be approached cautiously. Further testing across additional stores and longer observation periods may help determine the conditions under which the layout performs most effectively.

Metrics & Hypotheses

Evaluation Metrics



Sales Revenue



Number of Customers



Transactions per Customer

Trial Period and Store



February 2019 - April 2019



Store 77,86, and 88

H_0 : No Impact

- Sales revenue unchanged
- Customer count unchanged
- Transactions per customer unchanged

H_1 : Positive Impact

- Improvement in one or more key metrics
- Outperforms matched control stores

Control Store Matching Methodology

Objective

- Identify control stores that closely resemble each trial store before the intervention.

Similarity Scoring

- Correlation captures trend alignment
- Normalized RMSE captures magnitude alignment
- Scores combined for each metric
- Final score = average across all metrics

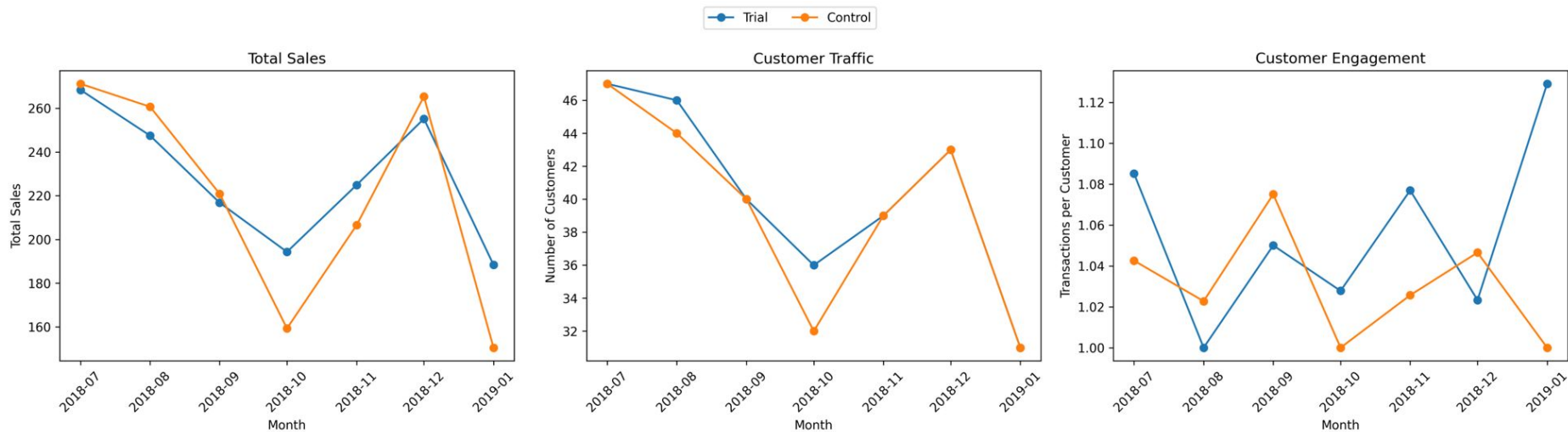
Matched Store Pairs

Trial Store	Matched Control	Similarity Score
77	233	0.757
86	138	0.816
88	201	0.691

Key message

Control stores were selected using a composite similarity score to ensure comparable pre-trial performance.

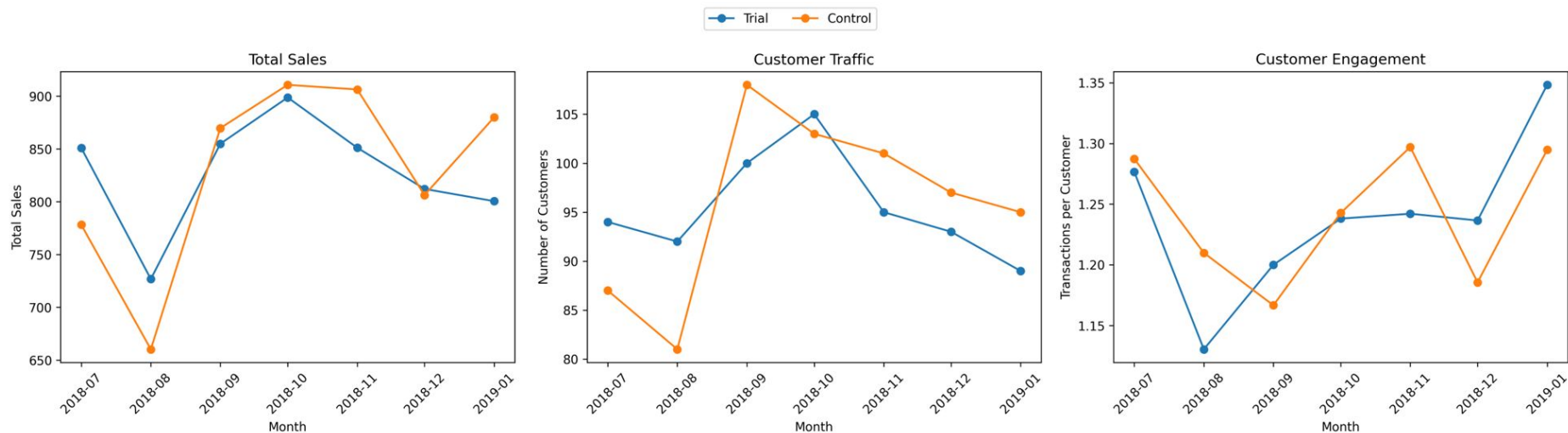
Pre-Trial Validation — Trial Store 77 vs Matched Control 233



Assessment

- Strong alignment in sales and customer traffic during the pre-trial period.
- Similar movement patterns across most weeks.
- Suitable control store for evaluating trial impact.

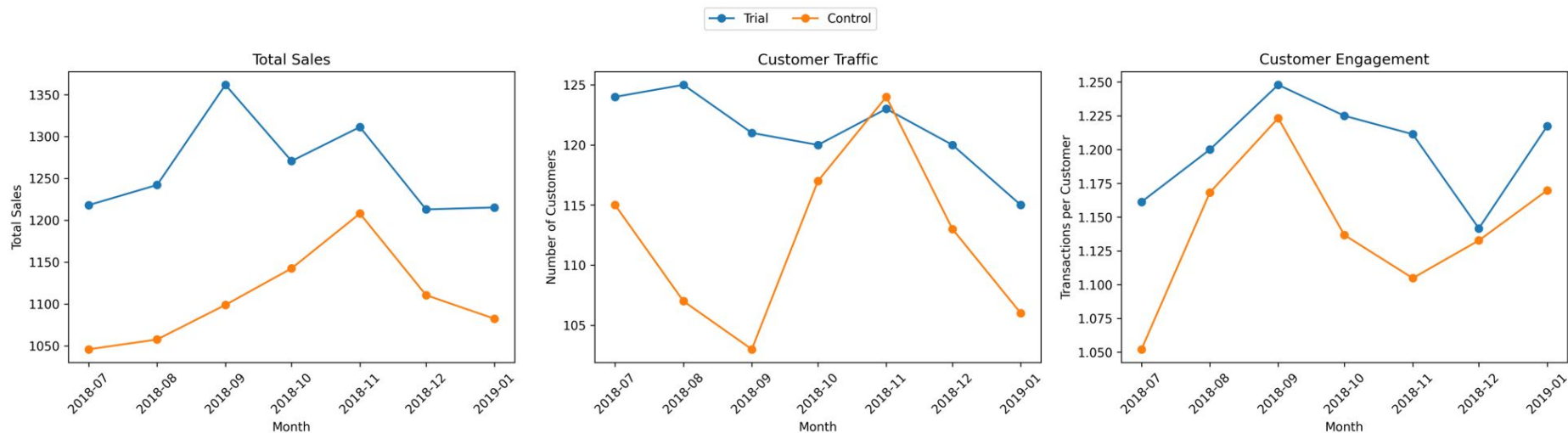
Pre-Trial Validation — Trial Store 86 vs Matched Control 138



Assessment

- High similarity across all three performance metrics.
- Consistent co-movement throughout the baseline period.
- Provides a robust benchmark for comparison.

Pre-Trial Validation — Trial Store 88 vs Matched Control 201



Assessment

- Moderate trend alignment across key performance metrics.
- Some differences in magnitude are observed between the trial and control stores.
- Overall comparability remains sufficient for trial evaluation.

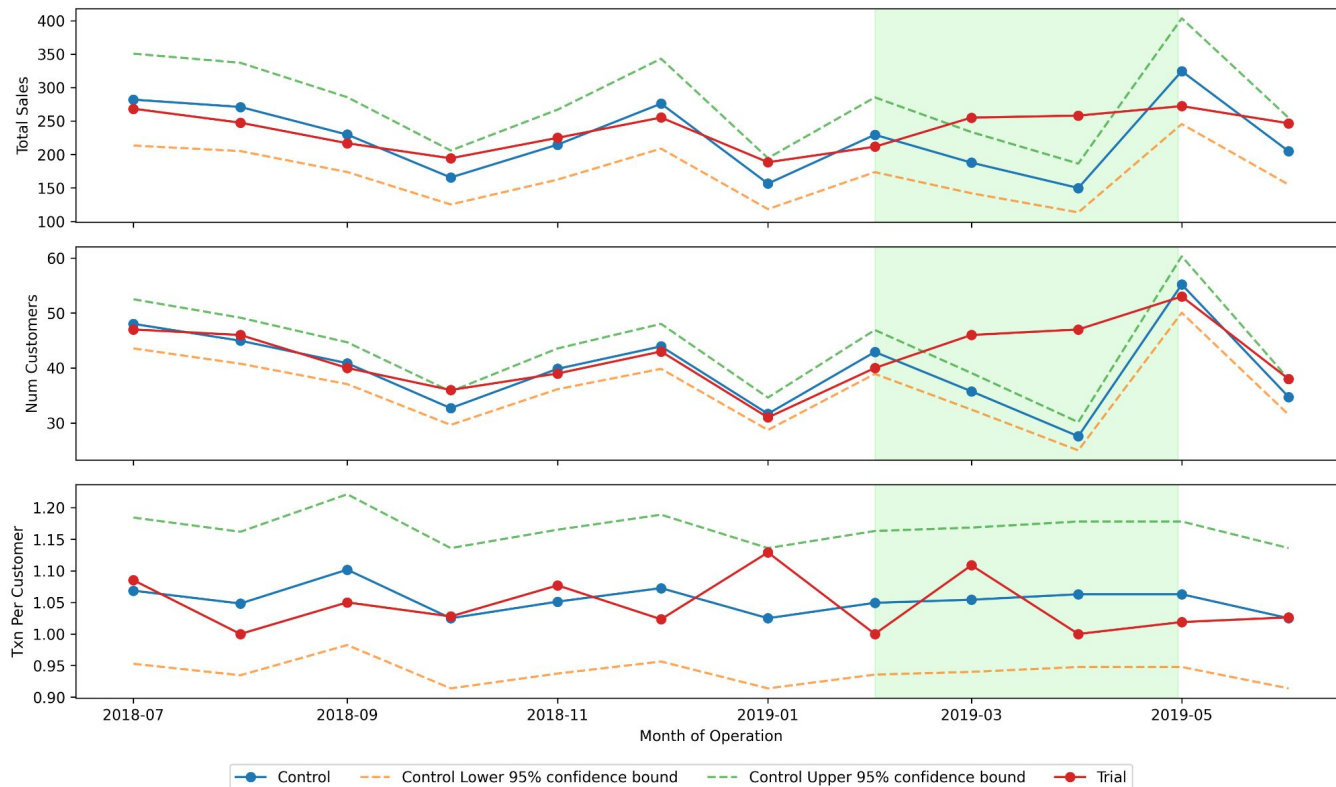
UPLIFT ANALYSIS

	Metric	Feb	Mar	Apr
STORE 77	Total sales	-7.78%	35.85%	72.14%
	Customer Traffic	-6.78%	28.63%	70.37%
	Customer Engagement	-4.69%	5.17%	-5.91%
STORE 86	Total sales	27.99%	6.66%	3.81%
	Customer Traffic	27.26%	6.51%	-0.40%
	Customer Engagement	0.88%	-8.31%	1.14%
STORE 88	Total sales	16.71%	-3.15%	-4.21%
	Customer Traffic	11.81%	-3.81%	-8.96%
	Customer Engagement	2.62%	3.82%	2.08%

Overall Asssesment

- **Impact varies across stores**, indicating that layout effectiveness is location-dependent.
- **Store 77 delivered the strongest results**, with significant gains in sales and customer traffic.
- **Store 86 showed moderate positive uplift**, but evidence was less consistent.
- **Store 88 produced mixed results**, with limited evidence of sustained improvement.
- **Customer traffic was the primary driver of uplift**, while customer engagement showed little change.

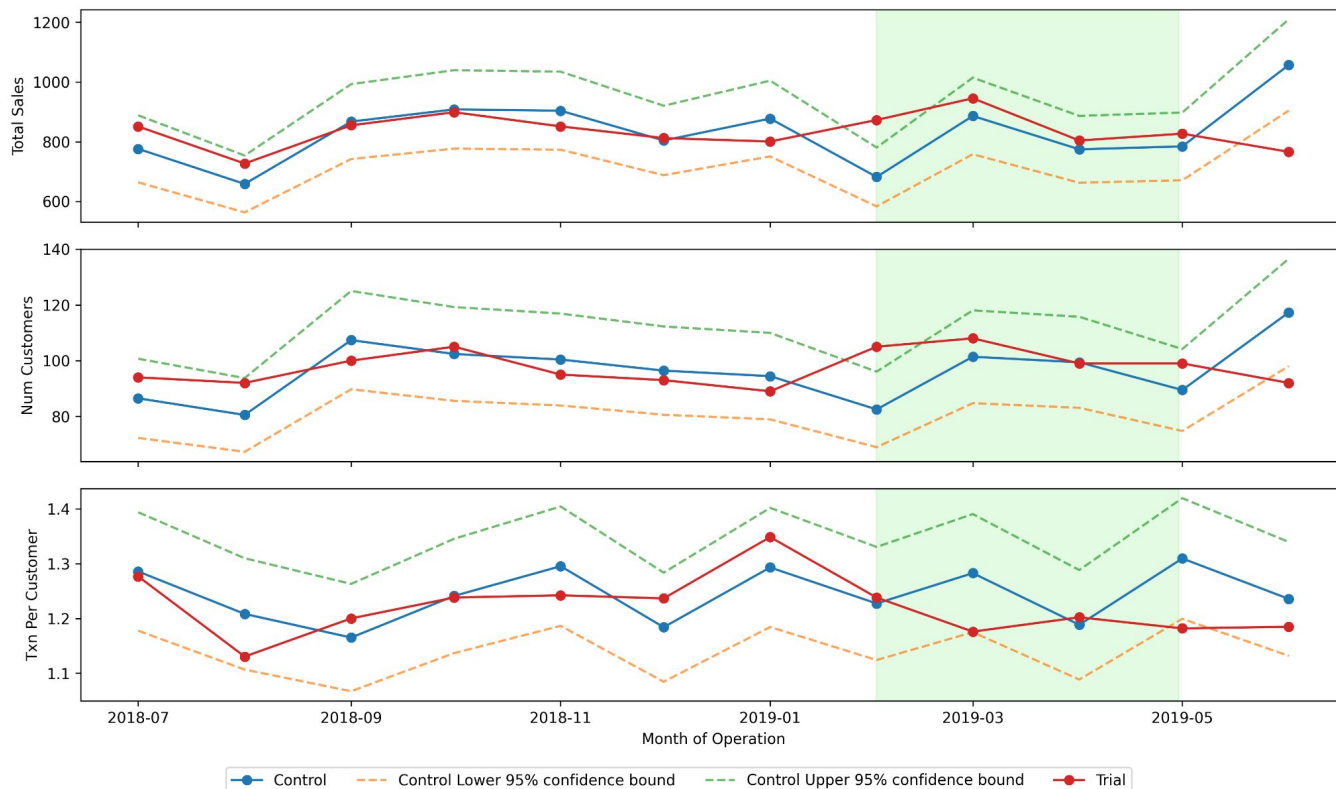
Significance of Uplift for Trial store 77



INSIGHTS

- Statistically significant uplift observed in March and April.
- Customer engagement remained within expected variation.

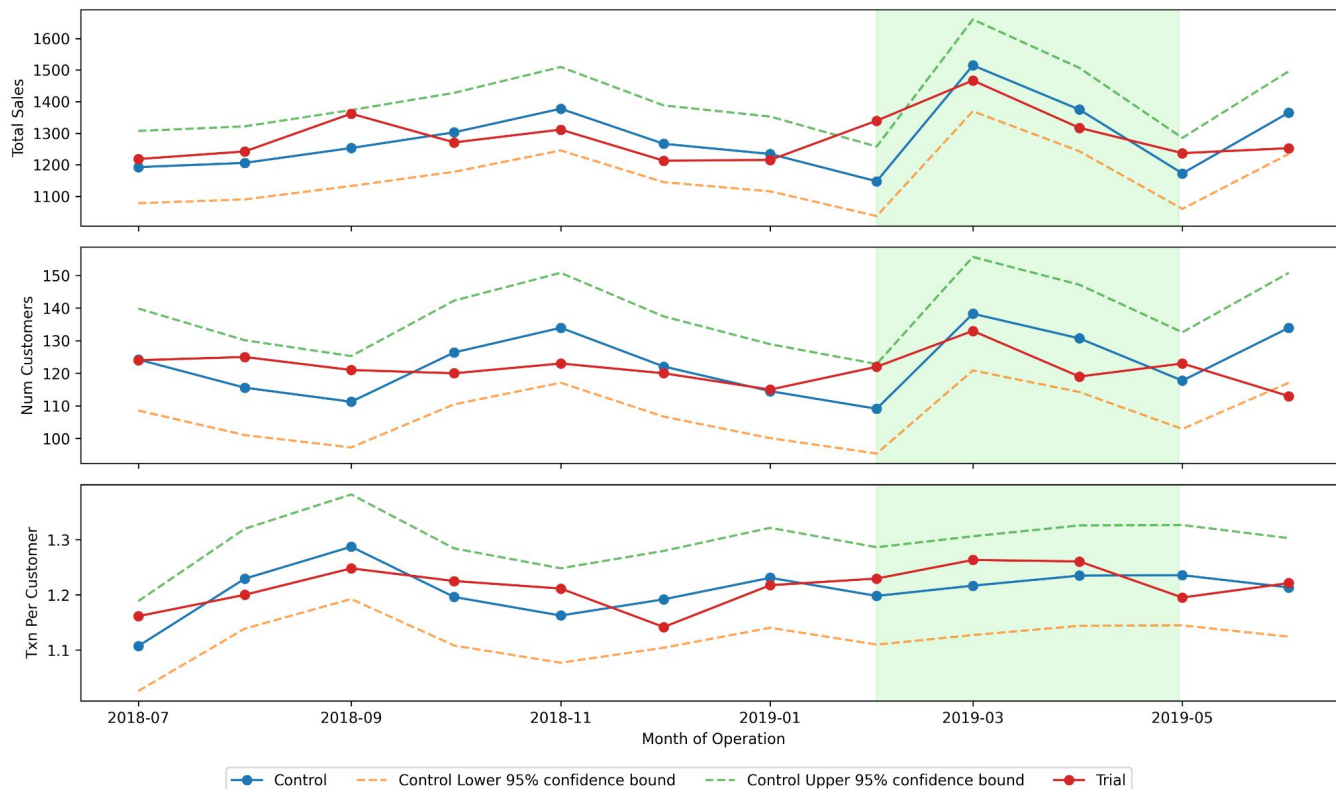
Significance of Uplift for Trial store 86



INSIGHT

Initial uplift was not sustained, with subsequent performance remaining within expected baseline variation.

Significance of Uplift for Trial store 86



INSIGHT

Performance remains largely within expected baseline variation.

THANK YOU